

HW Q i) $4x + y + 2z = 0, -3x + 2y + 4z = 0, 8x - y - 2z = 0$

ii) $x + 3y - 2z = 0, 2x - y + 9z = 0, x - 11y + 14z = 0$

Vectors

→ Any ordered n -tuple of numbers is called a vector. If $x_1, x_2, \dots, x_n$ are any n numbers then the ordered n -tuple $X = (x_1, x_2, \dots, x_n)$ is called an n -vector.

eg:- $(1, 0, 2, -1), (0, 1, 0, 0), (2, 2)$.

The n numbers $x_1, x_2, \dots, x_n$ are called components of n -vector.

→ A vector may be written as a row vector or as a column vector. If A is a matrix of order $m \times n$, then each row of A will be an n -vector and each column of A will be an m -vector.

→ A vector whose components are all zero is called zero vector, denoted by O . Thus $O = (0, 0, \dots, 0)$.

→ Let $X = (x_1, \dots, x_n)$ and $Y = (y_1, \dots, y_n)$ be 2 vectors. Then $X = Y$ iff $x_i = y_i, \forall i = 1, 2, \dots, n$ (Corresp components are equal).

→ $X + Y = (x_1 + y_1, \dots, x_n + y_n), KX = (Kx_1, \dots, Kx_n), K$ be a scalar.

Linear Dependence and Linear Independence of Vectors.

→ A set of n n -vectors $x_1, x_2, \dots, x_n$ is said to be linearly dependent if $\exists$ n scalars $K_1, K_2, \dots, K_n$ not all zero, s.t.

$$K_1 x_1 + K_2 x_2 + \dots + K_n x_n = 0.$$

→ A set of n n -vectors $x_1, x_2, \dots, x_n$ is said to be linearly independent if every relation of the type

$$K_1 x_1 + \dots + K_n x_n = 0 \Rightarrow K_1 = K_2 = \dots = K_n = 0.$$

Q. 8.7 The vectors $\alpha_1 = (1, 2, 4)$, $\alpha_2 = (2, -1, 3)$, $\alpha_3 = (0, 1, 2)$ and $\alpha_4 = (-3, 7, 2)$ are linearly dependent and find the relation b/w them?

Adding suitable multiples of α_1 to α_2 and α_4

$$\alpha_2 - 2\alpha_1 = (2, -1, 3) - (2, 4, 8) = (0, -5, -5)$$

$$(\alpha_4 + 3\alpha_1) = (-3, 7, 2) + (3, 6, 12) = (0, 13, 14)$$

Also, $\alpha_3 = (0, 1, 2)$

Adding suitable multiples of α_3 to the above vectors,

$$(\alpha_2 - 2\alpha_1) + 5\alpha_3 = (0, -5, -5) + (0, 5, 10) = (0, 0, 5)$$

$$(\alpha_4 + 3\alpha_1) - 13\alpha_3 = (0, 13, 14) - (0, 13, 26) = (0, 0, -12)$$

$$12(\alpha_2 - 2\alpha_1 + 5\alpha_3) + 5(\alpha_4 + 3\alpha_1 - 13\alpha_3)$$

$$12\alpha_2 - 24\alpha_1 + 60\alpha_3 + 5\alpha_4 + 15\alpha_1 - 60\alpha_3 = (0, 0, 60) + (0, 0, -60) = (0, 0, 0)$$

$$\Rightarrow 9\alpha_1 + 13\alpha_2 - 5\alpha_3 + 5\alpha_4 = (0, 0, 0)$$

$$\Rightarrow 9\alpha_1 - 12\alpha_2 + 5\alpha_3 - 5\alpha_4 = 0 \quad \text{--- (1)}$$

$\therefore \exists$ numbers $K_1 = 9, K_2 = -12, K_3 = 5, K_4 = -5$, which are not all zero s.t. $K_1\alpha_1 + K_2\alpha_2 + K_3\alpha_3 + K_4\alpha_4 = 0$

$\Rightarrow$ the vectors $\alpha_1, \alpha_2, \alpha_3$ are l.d.

Also (1) is the relation b/w them.

Linear Transformations.

$\rightarrow$ A relation $Y = AX$, where

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$AX = Y$$

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

f (cont'n), $a + f(n) + b + f(n)$

defines a linear transformation which carries any vector X into another vector Y over the matrix A which is called the linear operator of the transformation.

This transformation is called linear because $Y_1 = AX_1, Y_2 = AX_2$
 $\Rightarrow aY_1 + bY_2 = A(ax_1 + bx_2), \forall a, b.$

eg:- If $X = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$, $A = \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}$, then $Y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = AX = \begin{bmatrix} 5 \\ -5 \end{bmatrix}$

So that $(2, -3) \rightarrow (5, -5)$ under the transform defined by A .

→ If the transform matrix A is nonsingular i.e. if $|A| \neq 0$, then the linear transform is called non-singular or regular.

→ If the transform matrix A is singular i.e. if $|A| = 0$, then the L.T is called singular.

→ For a non-singular transform $Y = AX$, since A is non-singular A^{-1} exists and we can write the inverse transform which carries the vector Y back into the vector X as $X = A^{-1}Y$.

Orthogonal Transformation

The L.T $Y = AX$, where $Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$, $A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$

is said to be orthogonal if it transforms $y_1^2 + y_2^2 + \dots + y_n^2$ into $x_1^2 + x_2^2 + \dots + x_n^2$.

The matrix A of this transform is called an orthogonal matrix.

Now $X'X = [x_1 \dots x_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1^2 + \dots + x_n^2$

and $Y'Y = y_1^2 + \dots + y_n^2$

∴ If $Y = AX$ is an orthogonal transform, then,

$$X'X = x_1^2 + x_2^2 + \dots + x_n^2 = y_1^2 + \dots + y_n^2$$

$$= Y'Y = (AX)'(AX)$$

$$= (X'A')(AX)$$

$$(A'A) = I$$

$$X'X = X'(A'A)X$$

which holds only when $A'A = I$ or when $A'A = A^{-1}A$ (i.e. $A^{-1}A = I$)

or when $A' = A^{-1}$

→ A square matrix A is said to be orthogonal if $AA^T = A^T A = I$

Also for an orthogonal matrix $A^T = A^{-1}$

Q. S.T the transform $y_1 = 2x_1 + x_2 + x_3$, $y_2 = x_1 + x_2 + 2x_3$, $y_3 = x_1 - 2x_3$ is regular. Write down the inverse transform?

$$Y = AX$$

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}, A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 0 & -2 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$|A| = 2(-2-0) - 1(-2-2) + 1(0-1)$$

$$= -4 + 4 - 1 = -1 \neq 0$$

⇒ A is regular.

$$A^{-1} = \begin{bmatrix} 2 & -2 & -1 \\ -4 & 5 & 3 \\ 1 & -1 & -1 \end{bmatrix}$$

The inverse trans is given by $X = A^{-1}Y$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 & -2 & -1 \\ -4 & 5 & 3 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\Rightarrow x_1 = 2y_1 - 2y_2 - y_3, x_2 = -4y_1 + 5y_2 + 3y_3, x_3 = y_1 - y_2 - y_3$$

Q. P.T the following matrix is orthogonal $A = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$

$$A^T = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$$

$$AA^T = \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = I$$

Q. If A is an orthogonal matrix, P.T $|A| = \pm 1$

A orthogonal $\Rightarrow AA^T = I$

Q. P.T an inverse of an orthogonal matrix is orthogonal and its transpose is also orthogonal?

(i) A is orthogonal $\Rightarrow AA^T = I$

$$\Rightarrow (AA^T)^{-1} = I^{-1}$$

$$\Rightarrow (A^T)^{-1}(A)^{-1} = I$$

$$\Rightarrow (A^{-1})^T A^{-1} = I$$

$$\Rightarrow A^{-1} \text{ is orthogonal}$$

(ii) A is ortho $\Rightarrow AA^T = I$

$$\Rightarrow (AA^T)^T = I^T$$

$$\Rightarrow (A^T)^T A^T = I$$

$$\Rightarrow A^T \text{ is orthogonal}$$

Complex Matrices.

- If all the elements of a matrix are real nos, then it is called a real matrix or a matrix over $\mathbb{R}$.
- If at least one element of a matrix is a complex no. i.e., $a, b \in \mathbb{R}$, then the matrix is called a complex matrix.
- The matrix obtained by replacing the elements of a complex matrix A by the corresp conjugate complex nos is called the conjugate of the matrix A and is denoted by $\bar{A}$.

eg: $A = \begin{bmatrix} 2+3i & -7i \\ 5 & 1-i \end{bmatrix}$ then $\bar{A} = \begin{bmatrix} 2-3i & 7i \\ 5 & 1+i \end{bmatrix}$

$$\rightarrow (\bar{A}^t) = (\bar{A})^t = A^*$$

$\rightarrow$ A square matrix A is said to be Hermitian if $A^* = A$

$\rightarrow$ " " " " " Skew-Hermitian if $A^* = -A$.

$\rightarrow$ In a Hermitian matrix, the diagonal elements are real.

In a skew-Hermitian " " " " " zero or purely imaginary.

$\rightarrow$ A square matrix A is said to be unitary if $AA^* = A^*A = I$.

$\rightarrow$ The determinant of a unit matrix is of unit modulus.

The following results hold:-

i) $\overline{(\bar{A})} = A$ ii) $\overline{A+B} = \bar{A} + \bar{B}$ iii) $\overline{\lambda A} = \bar{\lambda} \bar{A}$ iv) $\overline{AB} = \bar{A} \cdot \bar{B}$
 v) $(A^*)^* = A$ vi) $(A+B)^* = A^* + B^*$ vii) $(\lambda A)^* = \bar{\lambda} A^*$ viii) $(AB)^* = B^* A^*$

Q. If $A = \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix}$, verify that A^*A is a Hermitian matrix?

$$A^t = \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix} \quad A^* = (\bar{A}^t) = \begin{bmatrix} 2-i & -5 \\ 3 & -i \\ -1-3i & 4+2i \end{bmatrix}$$

$$A^*A = \begin{bmatrix} 30 & 6-8i & -19-17i \\ 6+8i & 10 & -5+5i \\ -19+17i & -5-5i & 30 \end{bmatrix} = B$$

Now, $B^t = \begin{bmatrix} 30 & 6+8i & -19+17i \\ 6-8i & 10 & -5+5i \\ -19-17i & -5-5i & 30 \end{bmatrix}$ $B^* = (\bar{B}^t)$

$\therefore B = A^*A$ is Hermitian.

$= B$

Q. If A and B are Hermitian, s.t. $AB-BA$ is skew-hermitian?

$$A, B \text{ hermitian} \Rightarrow A^* = A, B^* = B$$

$$(AB-BA)^* = (AB)^* - (BA)^*$$

$$= B^* A^* - A^* B^*$$

$$= BA - AB$$

$$= -(AB-BA)$$

$\Rightarrow AB-BA$ is skew-hermitian

Q. S.T. if A is skew-hermi, iA is hermitian?

$$A \text{ skew hermi} \Rightarrow A^* = -A$$

$$(iA)^* = \bar{i} A^*$$

$$= -i A^*$$

$$= iA$$

$\Rightarrow iA$ is Hermitian

Q. S.T. every square matrix is expressible as the sum of a Hermitian and skew-hermitian matrix?

Let A be any square matrix.

$$(A+A^*)^* = A^* + (A^*)^* = A^* + A = A+A^* \Rightarrow A+A^* \text{ hermi}$$

$$(A-A^*)^* = A^* - (A^*)^* = A^* - A = -(A-A^*) \Rightarrow A-A^* \text{ skew hermi}$$

$$\text{Now, } A = \frac{1}{2}(A+A^*) + \frac{1}{2}(A-A^*) = P+Q.$$

where P is hermi, Q is skew hermi.

Q. If $N = \begin{bmatrix} 0 & 1+2i \\ -1+2i & 0 \end{bmatrix}$, obtain the matrix $(I-N)(I+N)^{-1}$ and s.t. it is unitary?

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1+2i \\ -1-2i & 1 \end{bmatrix}$$

$$|I+N| = 1 - (4i^2 - 1) = 6$$

$$\text{adj}(I+N) = \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix}$$

$$(I+N)^{-1} = \frac{1}{|I+N|} \text{adj}(I+N) = \frac{1}{6} \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix}$$

$$(I+N)(I-N)(I+N)^{-1} = \frac{1}{6} \begin{bmatrix} -4 & -2-4i \\ 2-4i & -4 \end{bmatrix} = A$$

$$A^T = \frac{1}{6} \begin{bmatrix} -4 & 2-4i \\ -2-4i & -4 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 1 & -1-2i \\ -1-2i & 1 \end{bmatrix} \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix}$$

$$A^* = \overline{(A^T)} = \frac{1}{6} \begin{bmatrix} -4 & 2+4i \\ -2+4i & -4 \end{bmatrix}$$

$$A^* A = \frac{1}{36} \begin{bmatrix} 36 & 0 \\ 0 & 36 \end{bmatrix} = I$$

$$\Rightarrow A = (I-N)(I+N)^{-1} \text{ is unitary}$$

Characteristic Equation

If A_n is any square matrix, we can form the matrix $A - \lambda I$, where λ is a scalar and I_n is the unit matrix. The determinant of this matrix is equal to zero, is called the characteristic eqn of A .

$$\text{i.e., } |A - \lambda I| = \begin{vmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} - \lambda & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{vmatrix} = 0$$

→ on expanding the chara eqn can be written as a polynomial eqn of deg n in λ of the form

$$(-1)^n \lambda^n + K_1 \lambda^{n-1} + K_2 \lambda^{n-2} + \dots + K_n = 0$$

→ The roots of this eqn are called the chara roots or latent roots or eigen values of A.

→ The sum of e-values of A is equal to trace of A.
(Sum of diag elements = trace of A).

Eigen Vectors (Tent)

Q. Find the e-vec and e-values of the matrix $A = \begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$?

$$|A - \lambda I| = 0 \quad \left| \begin{array}{cc} 1-\lambda & -2 \\ -5 & 4-\lambda \end{array} \right| = 0$$

$$(1-\lambda)(4-\lambda) - 10 = 0$$

$$\lambda^2 - 5\lambda - 6 = 0$$

$$\lambda = 6, -1.$$

Corresponding to $\lambda = 6$, the e-vectors are given by $(A - 6I)X = 0$.

$$\begin{bmatrix} 1-6 & -2 \\ -5 & 4-6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} -5 & -2 \\ -5 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0.$$

$$\begin{matrix} -5 & -2 \\ -5 & -2 \end{matrix}$$

We get only one independent eqn $-5x_1 - 2x_2 = 0$

$$\Rightarrow \frac{x_1}{2} = \frac{x_2}{-5} = K_1$$

$$\Rightarrow \text{eigen vector is } K_1 \begin{bmatrix} 2 \\ -5 \end{bmatrix}$$

$$\frac{x_1}{2} = \frac{x_2}{-5} = K$$

$$x_1 = 2K$$

$$x_2 = -5K$$

Correspo $\lambda = -1$, the e-vector are given by $\begin{bmatrix} 2 & -2 \\ -5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

We get only one inde eqn $2x_1 - 2x_2 = 0$.

$$\Rightarrow x_1 = x_2 = K_2 \text{ giving the e. vect } K_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Q. Find the e.va and e.ve of the matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$?

chara eqn $\lambda^3 + \lambda^2 - 21\lambda - 45 = 0$

$(\lambda+3)(\lambda^2 - 2\lambda - 15) = 0$

$(\lambda+3)(\lambda+3)(\lambda-5) = 0$

$\lambda = -3, -3, 5$

corresp $\lambda = -3$ e. vec are $(A+3I)x = 0$

$$\begin{bmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & -\lambda \end{bmatrix} = 0$$

$(-2-\lambda)(-\lambda+12) - 2$

$(-2\lambda-6) - 3(-4+12\lambda) = 0$

$2\lambda - 2\lambda^2 + 24 + \lambda^2 - \lambda^3 + 12\lambda$

$+4\lambda - 12 + 12 + 3 - 3\lambda = 0$

$-\lambda^3 - \lambda^2$

choosing

$\lambda_2 = 0$

$\lambda_1 + 2\lambda_2 - 3\lambda_3 = 0$

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = 0$$

$\Rightarrow \frac{\lambda_1}{3} = \frac{\lambda_2}{0} = \frac{\lambda_3}{1} = k_1$

$\Rightarrow$ e. vec is $k_1 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$

$\frac{\lambda_1}{3} = \lambda_3$

$\frac{\lambda_1}{0} = \frac{\lambda_2}{3} = \frac{\lambda_3}{2} = k$

choosing $\lambda_3 = 0$, $\lambda_1 + 2\lambda_2 = 0$

$\therefore \frac{\lambda_1}{2} = \frac{\lambda_2}{-1} = \frac{\lambda_3}{0} = k_2 \Rightarrow$ e. vec is $k_2 \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$

Any other e. vec corresp $\lambda = -3$ will be a linear combi of these 2.

$\lambda = 5$, e. vec are given by $\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = 0$

Since only 2 of them are indep., we can omit one of them.

From 1st 2 eqns,

$\frac{\lambda_1}{-12-12} = \frac{\lambda_2}{-6-42} = \frac{\lambda_3}{28-4}$
 $\frac{\lambda_1}{-24} = \frac{\lambda_2}{-48} = \frac{\lambda_3}{24}$
 $\frac{\lambda_1}{1} = \frac{\lambda_2}{2} = \frac{\lambda_3}{-1} = k_2$

$-7\lambda_1 + 2\lambda_2 - 3\lambda_3 = 0$

$2\lambda_1 - 4\lambda_2 - 6\lambda_3 = 0$

$$\begin{bmatrix} -1 & -2 & -5 \\ 2 & -4 & -6 \\ -7 & 2 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 5 \\ 2 & -4 & -6 \\ -7 & 2 & -3 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 5 \\ 0 & -8 & -16 \\ 0 & 16 & 32 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow$ e. vec $k_2 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

Q. S.T if $\lambda_1, \dots, \lambda_n$ are the latent roots of A , then A^3 has latent roots $\lambda_1^3, \dots, \lambda_n^3$.

Let λ be a latent root of A .
 $(A - \lambda I)x = 0$

$$Ax = \lambda x$$

$$\Rightarrow A^2(Ax) = A^2(\lambda x)$$

$$\Rightarrow A^3x = \lambda(A^2x) \quad \text{--- ①}$$

$$\text{But } A^2x = A(Ax) = A(\lambda x)$$

$$= \lambda(Ax)$$

$$= \lambda \lambda x$$

$$= \lambda^2 x$$

$$\therefore \text{①} \Rightarrow A^3x = \lambda(\lambda^2 x) = \lambda^3 x$$

$$\Rightarrow \lambda^3 \text{ is a latent root of } A^3.$$

$\therefore$ if $\lambda_1, \dots, \lambda_n$ are latent roots of $A \Rightarrow \lambda_1^3, \dots, \lambda_n^3$ are latent roots of A^3 .

Q. If λ be an e.v.a of a non-sing matrix A , S.T

i) λ^{-1} is an e.v.a of A^{-1}

ii) $\frac{|A|}{\lambda}$ is an e.v.a of $\text{adj } A$.

i) λ is an e.v.a of $A \Rightarrow Ax = \lambda x, x \neq 0$

$$x = A^{-1}(\lambda x)$$

$$\Rightarrow x = \lambda(A^{-1}x)$$

$$\frac{1}{\lambda} x = A^{-1}x$$

$$\Rightarrow A^{-1}x = \frac{1}{\lambda} x$$

$\Rightarrow \frac{1}{\lambda}$ is an e.v.a of A^{-1}

ii) $Ax = \lambda x, x \neq 0$

$$(\text{adj } A) Ax = (\text{adj } A) \lambda x$$

$$(\text{adj } A) A = \lambda (\text{adj } A) X$$

$$|A| \cdot I X = (\lambda (\text{adj } A) X)$$

$$\frac{|A|}{\lambda} X = (\text{adj } A) X$$

$$\Rightarrow (\text{adj } A) X = \frac{|A|}{\lambda} X$$

$$\Rightarrow \frac{|A|}{\lambda} \text{ is an e.v.a of adj } A$$

Cayley Hamilton Theorem

Every square matrix satisfies its own chara eqn.

i.e., if the chara eqn of the n^{th} order square matrix A is

$$|A - \lambda I| = (-\lambda)^n + k_1 \lambda^{n-1} + \dots + k_n = 0$$

$$\text{then } (-1)^n A^n + k_1 A^{n-1} + \dots + k_n I = 0$$

Q. Verify Cayley-Hamilton thm for $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$. & compute A^{-1}

$$|A - \lambda I| = 0 \Rightarrow \lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$

To verify Cayley-Hamilton thm, we've to s.t

$$A^3 - 6A^2 + 9A - 4I = 0, \quad \text{--- (1)}$$

$$\text{Now, } A^2 = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}, \quad A^3 = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$$

$$\therefore A^3 - 6A^2 + 9A - 4I = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - 6 \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} + 9 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= 0$$

$$0 \times A^{-1} \Rightarrow A^2 - 6A + 9I - 4A^{-1} = 0$$

$$\Rightarrow A^{-1} = \frac{1}{4} (A^2 - 6A + 9I)$$

$$= \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

Q. Find the chara eqn of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence find the matrix represented by $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$

$$|A - \lambda I| = 0 \Rightarrow \lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

$$\text{By C-H thm, } A^3 - 5A^2 + 7A - 3I = 0$$

$$\text{Now, } A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$$

$$= A^5(A^3 - 5A^2 + 7A - 3I) + A(A^3 - 5A^2 + 7A - 3I) + (A^2 + A + I)$$

$$= A^5 \cdot 0 + A \cdot 0 + (A^2 + A + I)$$

$$= A^2 + A + I$$

$$= \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$$